

Element L: Presentation of Designer's Recommendations

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students' ability to rest.

The insects make the space noisy and affect the students' ability to rest. For students who will design insect catchers like this, we suggest they first evaluate the amount of space where the insect catcher will go. You should observe where insects usually are in the room. Note where they enter and if there are things like pets or children around. Also, think about where you can place the insect catcher easily. Knowing the space well will help you decide the size and shape your insect catcher should be. To ensure your insect catcher helps, talk with the people who have the insect problem. We talked to student E., asking them what bothers them most about the insects and how they would like to get rid of them. Show them your early ideas and listen to what they say. Their thoughts will make your design much better and more useful. Communicating often with your team online can also help you all stay on the same page, especially when you are working on things together, from different locations.

We also think it is important to estimate how long each part of the project will take, including building and moving things. This ensures that you and your team will aim to get things done on or before a certain deadline. If someone else might build your design, make sure your plans are very clear and easy to follow. Write down how to test it, how to build it, and what materials you need. To keep the work area tidy, plan to build the insect catcher in pieces and put it together where it will be used. This way, you will not make a big mess or have trouble carrying it. When you are just starting to think about ideas, make quick models out of cheap things like cardboard or foam. You can also use a simple three-dimensional printer if you have one. These

easy models can help you see if the size and shape are right and how the basic catching part might work. For example, you could make a simple box with sticky paper inside to see if it is a good size to catch insects. Once you have a good idea, build models of the main parts that catch the insects. If you have a special trap or opening, make a model of just that part. Test it to see if it actually catches insects well and if it breaks easily. This will help you make those parts better. Show your simple models to the people with the insect problem and ask them what they think. Observe how they interact with it and get their ideas on how to make it easier to use and more effective. Getting their hands-on feedback is super helpful.